recognize the antigen, produce antibody against it and eventually produce memory lymphocytes. The primary immune response is slow and short-lived.

Secondary immune response

The secondary immune response occurs when a person is exposed to the same antigen again. During this time, immunological memory has been established and the immune system can start producing antibodies **immediately**. Within hours after recognition of the antigen, a new army of plasma cells are generated. Within 2 to 3 days, the antibody concentration in the blood **rises steeply** to reach much higher level than primary response. This is also called as "**booster response**".

7.3.4 Lymphoid organs

Immune system of an organism consists of several structurally and functionally different organs and tissues that are widely dispersed in the body. The organs involved in the origin, maturation and proliferation of lymphocytes are called **lymphoid organs (Fig. 7.11).** Based on their functions, they are classified into **primary or central lymphoid organs** and **secondary or peripheral lymphoid organs**. The primary lymphoid organs provide appropriate environment for lymphocytic maturation. The secondary lymphoid organs trap antigens and make it available for mature lymphocytes, which can effectively fight against these antigens.

Primary lymphoid organs

Bursa of Fabricius of birds, bone marrow and thymus gland of mammals constitute the primary lymphoid organs involved in the production and early selection of lymphocytes. These lymphocytes become dedicated to a particular **antigenic specificity**. Only when the lymphocytes mature in the primary lymphoidal organs, they become **immunocompetent cells**. In mammals, B cell maturation occurs in the bone marrow and T cells maturation occurs in the thymus.

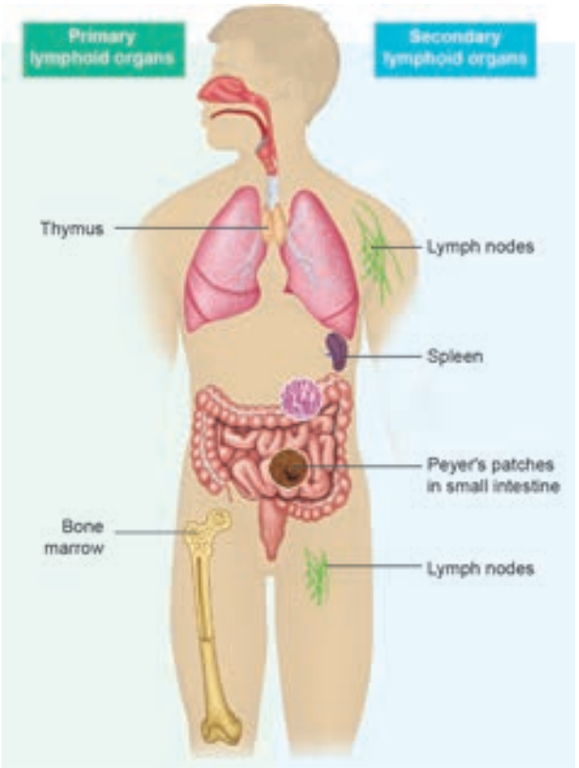
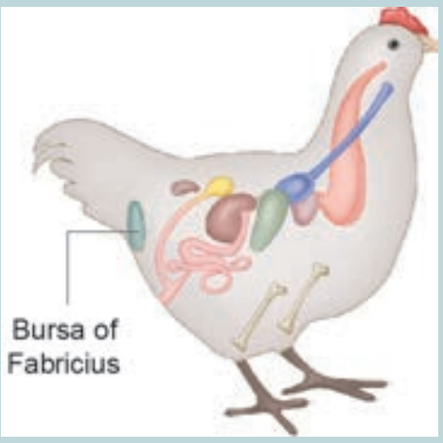


Fig. 7.11 Lymphoid organs in human body



Bursa of Fabricius is a primary lymphoid organ of birds. It is attached to the dorsal side of the cloaca. B lymphocytes mature in the bursa and bring about humoral immunity.

Thymus

The thymus is a flat and bilobed organ located behind the sternun, above the heart. Each lobe of the thymus contains numerous lobules, separated from each other by connective tissue called septa. Each lobule

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